

Glaucoma Assessment from Fundus Images with Fundus to OCT Feature Space Mapping

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Early detection and treatment of glaucoma is of interest as it is a chronic eye disease leading to an irreversible loss of vision. Existing automated systems rely largely on fundus images for assessment of glaucoma due to their fast acquisition and cost-effectiveness. **Optical Coherence Tomographic (OCT)** images provide vital and unambiguous information about nerve fiber loss and optic cup morphology, which are essential for disease assessment. However, the high cost of OCT is a deterrent for deployment in screening at large scale. In this article, we present a novel CAD solution wherein both OCT and fundus modality images are leveraged to learn a model that can perform a mapping of fundus to OCT feature space. We show how this model can be subsequently used to detect glaucoma given an image from only one modality (fundus). The proposed model has been validated extensively on four public and two private datasets. It attained an AUC/Sensitivity value of 0.9429/0.9044 on a diverse set of 568 images, which is superior to the figures obtained by a model that is trained only on fundus features. Cross-validation was also done on nearly 1,600 images drawn from a private (OD-centric) and a public (macula-centric) dataset and the proposed model was found to outperform the state-of-the-art method by 8% (public) to 18% (private). Thus, we conclude that fundus to OCT feature space mapping is an attractive option for glaucoma detection.

CCS Concepts: • **Applied computing** → **Health care information systems**; *Health informatics*; • **Computing methodologies** → **Neural networks**; *3D imaging*;

Additional Key Words and Phrases: Glaucoma assessment, fundus, optical coherence tomography, deep learning, GAN

ACM Reference format:

Divya Jyothi Gaddipati and Jayanthi Sivaswamy. 2021. Glaucoma Assessment from Fundus Images with Fundus to OCT Feature Space Mapping. *ACM Trans. Comput. Healthcare* 3, 1, Article 10 (October 2021), 15 pages.
<https://doi.org/10.1145/3470979>

1 INTRODUCTION

Damage to the optic nerve leads to glaucoma, a major eye disease which gradually leads to an irreversible loss in vision. Symptoms include elevated **intra-ocular pressure (IOP)**, thinning of the **retinal nerve fiber layer (RNFL)**, and an enlargement of **optic cup (OC)** often referred to as “cupping.” Cupping is quantitatively assessed using the vertical **cup-disc diameter ratio (CDR)** and the thickness profile of the rim, the region between the OC and the **optic disc (OD)**. Early detection of glaucoma is required to prevent degradation of the optic nerves and arrest the progression of the disease.

A majority of the **computer-aided diagnostic (CAD)** solutions proposed for detecting glaucoma are based on fundus camera images given their cost effectiveness and wider availability. Most of these solutions rely on

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2637-8051/2021/10-ART10 \$15.00

<https://doi.org/10.1145/3470979>

two distinct approaches: analysis of the OD region, with or without segmentation of the OD and OC. In the first approach, handcrafted features are extracted and then classified by various machine learning strategies. However, advances in neural networks have enabled learning the features using deep **convolutional neural network (CNN)** architectures [1, 2]. Variants of approaches can be found in recent literature [3]. In [4], a novel encoder-decoder-based design is proposed where the encoder is used to obtain a latent-space distribution which is then used by a classifier for glaucoma detection. Transfer learning with popular architectures like VGG-16, InceptionNet, and ResNet50 have also been explored [5]. Image synthesis using a Semi-Supervised **Deep Convolutional Generative Adversarial Network (DCGAN)** is another novel attempt to generate labeled retinal images cropped around the OD. Such a network has been trained on a large dataset (more than 80,000 images), to design a semi-supervised classifier for glaucoma assessment in [6] and the reported Sens./Spec./AUC are 0.8290/0.7986/0.9017. Another approach based on transfer learning/fine-tuning is reported in [7]. VGG16, ResNet, and GoogleNet have been experimented with and without transfer learning on a combined dataset of around 1,700 images that was split into two sets for training and testing. The highest performance was found to be obtained using a fine-tuned VGG16 whose BAcc./Sens./Spec./AUC are 0.88/0.87/0.89/0.94.

The second approach is based on parametric analysis using CDR and rim profile, which are also used in the clinical evaluation process. This requires the segmentation of OD and OC. Segmentation solutions that have been proposed range from super-pixel classification [8, 9], weighted reconstruction of OD and OC [10], and depth-based joint estimation [11]. The U-net is a neural network architecture proposed for deep learning-based segmentation. Adapting this has also been attempted for glaucoma assessment, either with very little [12] or a more extensively modified form [13]. The framework proposed by [13], M-Net is a pixel-wise classification-based approach which first classifies each pixel as OD or OC or non-target, and then uses ellipse-fitting to approximate the boundaries and then CDR was extracted for glaucoma detection. While the OD segmentation methods report very high accuracy, cup segmentation is still an open problem and errors in this stage has a bearing on the final diagnosis. A few approaches have combined both segmentation- and image-based features for glaucoma classification by fusing them in various ways. For instance, [14] proposed a framework that fuses both segmentation-based and image-based features within a co-training-based semi-supervised setting to classify the input. Another framework, DE-Net, introduced by [15], integrates four independent neural network streams using U-Net and ResNet-50 whose predictions are combined to obtain the final screening result.

However, **Optical Coherence Technology (OCT)** provides a detailed cross-sectional view of the retinal tissues. It has advantages over fundus imaging with regard to the assessment of glaucoma. Many clinical studies have shown OCT to be more sensitive compared to fundus photography for assessing glaucoma [16]. Most of the existing work on OCT-based assessment is based on extracting features such as discrete wavelet coefficients [17], from the segmented retinal structures and classifying them using machine learning techniques. More recently, DeepMind introduced a framework [18], which is an ensemble of five segmentation and five classification networks, that analyzes OCT scans for both the tasks, and reported a performance on par with the clinical experts. They have demonstrated that their system offers potential usage in a clinical setting.

Thus, in the prevailing scenario, fundus or OCT image analysis is typically used separately for automated assessment of glaucoma, whereas including information from multiple modalities, which provide complementary information, would be of interest and may have the potential for a more accurate diagnosis. However, this has not been explored to its full potential yet.

Fundus imaging is less expensive and more widely available compared to OCT imaging, which implies that it might not be feasible to have a combination of modalities available in a glaucoma screening scenario. We therefore look at the problem of glaucoma diagnosis anew, keeping this constraint in mind and pose it as a missing data problem. In other words, the CAD solution aims to learn a model that leverages information from fundus and OCT images during a training phase and perform an accurate assessment during a subsequent online detection phase in the absence of OCT data. This extends the possibilities of deploying the CAD systems where the OCT devices are too expensive for widespread usage. The main novelty in the proposed solution lies in

this aspect of combining the imaging modalities which provide complementary information during training, and deriving the screening result using just the fundus modality during testing. The strategy we adopt is to learn a model that maps fundus to an OCT-feature space such that given a fundus image, the model is able to generate additional features from this space. The strength of the proposed method is the training does not rely on paired fundus-OCT data. Finally, to provide a more accurate diagnosis, a model is proposed which generates the aforementioned fundus-to-OCT mapped features along with the fundus-specific features which are fused and used for classifying the input as normal or glaucomatous. We also demonstrate that our approach has a significant improvement in classification accuracy over that obtained from a corresponding unimodal approach.

2 METHODOLOGY

When a structure is imaged with two different modalities, the information captured across the two may be different (even complementary) but statistically or otherwise related. This is true even if one is a functional imaging modality while the other is a structural imaging modality. This is at the heart of the use of mutual information for cross modal registration. In the problem at hand, both OCT and fundus imaging are structural imaging modalities with one representing the retinal surface while the other its cross section. Despite this difference, it should be possible to relate the two in a feature space. Further, it is desirable that this relation is defined (or learned) at a semantic level (normal versus glaucomatous class) as the abstraction will then be agnostic to variations within the classes and across individual eyes (retina). Thus, the OCT and fundus images need not be paired, a key advantage as sourcing paired data is more difficult. Hence, in our design, we propose to use a network to derive features from labeled OCT volumes and a second network to perform an adversarial-learning-based mapping between features derived from labeled fundus images to the OCT features. It is notable that this is different from the domain adaptation technique wherein both the images are mapped to a common feature space. Finally, we design a classification network that fuses features and performs the glaucoma assessment.

2.1 Overview of the Proposed Method

The proposed framework consists of three modules: (i) preprocessing of OCT volumes and fundus images as described in Section 2.2; (ii) establishing the OCT-feature space to which the fundus images are mapped using a generative adversarial technique as described in Section 2.3; and (iii) an end-to-end model for deriving the decision on input fundus images. This is described in Section 2.4.

2.2 Preprocessing

2.2.1 OCT Volume. Preprocessing of OCT is done for three main tasks: denoising, retinal curvature flattening, and **region of interest (ROI)** extraction. OCT volumes are generally corrupted by speckle noise whose characteristics depend on the retinal tissue and scanner type. Denoising of every OCT volume was done using speckle reducing anisotropic diffusion [19], followed by bilateral filtering to obtain a smooth image while preserving the edges. The retinal region has a natural curvature that can vary both across volumes and within each volume. Hence, flattening was performed in two steps using a method similar to that introduced in [20]. First, for every slice (or B-scan), a rough estimate of the outer **Retinal Pigment Epithelium (RPE)** layer boundary is obtained, which is a 1D vector consisting of the position of the brightest pixel in each column. As the OCT data does not contain any other retinal diseases that might affect the layers, the brightest pixel can be considered for obtaining the required boundary. Next, a quadratic polynomial is fit to the resultant vector after removing the outlier pixels, which is then mapped to a straight line by shifting the columns. Next, all the slices are aligned successively across the volume by an exhaustive search of the axial translations of pixels which maximize the normalized cross correlation between the adjacent slices. The third task of extraction of the ROI from an OCT volume serves to reduce computations since the layer region only occupies 30–40% in each slice. It was performed using the method in [21]. Each slice was projected along the columns to obtain a 1D projection and a Gaussian function is fitted on the same. The mean value of the resultant represents the row index of the center

of the region containing the layers. Finally, every slice is cropped such that the resultant region is 200 pixels in height and 304 pixels in width. The number of slices in each volume is reduced by one-third to further reduce the computational burden during training.

2.2.2 Fundus Image. Fundus images often suffer from non-uniform illumination. This was corrected by luminosity and contrast normalization [22] as follows. For each pixel ($I(x, y)$), the mean ($\mu(x, y)$) and standard deviation ($\sigma(x, y)$) are computed within a fixed-size window of size 512×512 . These statistics are used to identify the background pixels using the Mahalanobis distance $D(x, y)$ and finally the corrected image $I_c(x, y)$. D and I_c are computed as

$$D(x, y) = \left| \frac{I(x, y) - \mu(x, y)}{\sigma(x, y) - \sigma_\epsilon} \right|; \quad I_c(x, y) = \left| \frac{I(x, y) - \mu_B(x, y)}{\sigma_B(x, y) - \sigma_\epsilon} \right|, \quad (1)$$

where $\mu_B(x, y)$ and $\sigma_B(x, y)$ are the mean and the standard deviation of the background pixels; a pixel I is defined to belong to the background if $D(x, y) < 1.3$.

Since we are interested only in the optic nerve head and the peripapillary regions, a region of interest enclosing these was extracted from the given fundus image as follows. Firstly, the center of the OD is determined using a Hough transform-based OD localization method [23]. Next, a region of 304 pixels in height and width is cropped around this center.

2.3 GAN-Based Mapping of Fundus to OCT Feature Space

2.3.1 Establishing the OCT-Feature Space. A Capsule network [24] is employed for generating the high-level features from OCT. It helps to overcome the issues faced in a CNN like requirement of large datasets incorporating full variations of the input and handling imbalanced data. Limited annotated data availability and class imbalance are common in glaucoma assessment, especially in the case of OCT. Recent studies [25, 26] have demonstrated that Capsule networks have the potential to obtain highly discriminative features even when trained with far less data and are more robust to an imbalanced class distribution.

The existing architecture of the Capsule network cannot handle 3D data and hence modifications were made to handle the OCT volumes. The final network is simple and shallow consisting of three blocks of 3D Conv-BatchNorm-LeakyRelu-SpatialDropout layers ending with dynamic routing layers as shown in Figure 1(A). The 3D-Conv blocks are added before the routing layers to shrink the volume and reduce the computational overhead during training. The features from the last 3D-Conv block are used as inputs to the primary capsules. The primary capsule layer performs 256 (32×8) convolutional operations and produces 32 capsule maps; each capsule being an 8D vector. Since the aim is to classify OCT volumes as normal or glaucomatous, the class capsule layer is designed to have two capsules. Each class is represented as a 16D capsule, where the length of the capsule determines the probability of the input belonging to a particular class. Each of these capsules receives input from all capsules in the primary capsule layer. An iterative dynamic routing algorithm is employed to achieve the interaction between the primary and class capsule layers, determining the passage of information between them. The weights or coupling coefficients between these layers, determine the influence that a particular capsule has on another capsule in the layer above and thus, determine how information flows between the layers. A non-linear squashing function (S) [24] is applied to the output vector of the class capsule layer, which ensures that the orientation of the vector is unchanged, while the length is normalized between 0 and 1. This indicates the probability that the entity represented by the capsule exists in the current input.

The network is trained with joint supervision of two loss functions: margin loss and center loss.

$$L = \lambda_1 L_m + \lambda_2 L_c \text{ where } \lambda_1 = 1, \lambda_2 = 0.3. \quad (2)$$

L_m is the margin loss defined in a similar manner as [24], which is computed for each capsule. It is defined as

$$L_m = T_k \max(0, m^+ - \|v_k\|)^2 + (1 - T_k) \max(0, \|v_k\| - m^-)^2, \quad (3)$$

where $T_k = 1$ if and only if an input of class k is present and $m^+ = 0.9$ and $m^- = 0.1$.

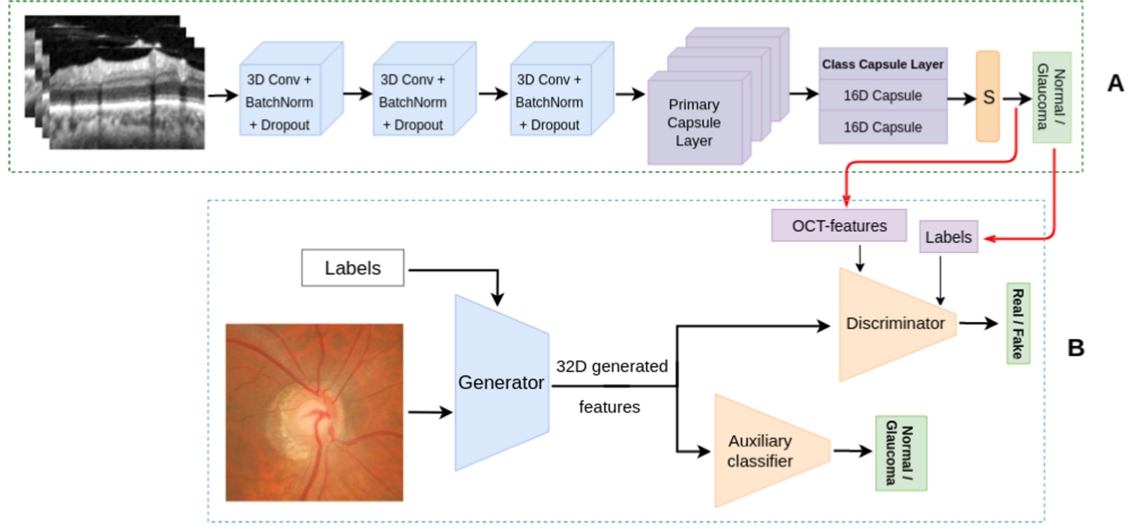


Fig. 1. Proposed two-stage system for mapping fundus images to the OCT-feature space. Both networks are trained separately. (A) Capsule network generating the OCT-features. (B) Mapping network based on a GAN model.

L_c is the center loss which aids learning the centers for each class simultaneously and penalizes the distances between the deep features learned and their corresponding class centers [27]. It is defined as

$$L_c = \frac{1}{2} \sum_{i=1}^m \|x_i - c_{y_i}\|_2^2, \quad (4)$$

where x_i denotes the i^{th} deep feature obtained as the output of the class capsule layer; c_{y_i} denotes the center belonging to the y_i^{th} class, which is updated for each training batch and m is the batch size. For each of the OCT images, a feature vector is obtained after the class capsule layer, which is used in the next stage.

2.3.2 Mapping the Fundus to OCT-Feature Space Using GAN. Although highly discriminative features are learned from OCT, operating on such high dimensional data has high computational and time complexity. To avoid this, we propose a strategy to transfer the generalization ability of the Capsule network into a smaller model which operates on fundus. This is done using the capsule features generated, since they represent the valuable information of OCT in a lower dimensional space. For this purpose, we make use of an adversarial network to learn a mapping from fundus to the OCT-feature space.

Generative Adversarial Networks (GANs), introduced by [28], are generative models that learn a mapping from random noise vector to output. **Conditional GAN (CGAN)**, introduced by [29], extends the GAN model allowing the generation of samples with certain attributes. It learns mapping from a random noise vector z and observed input x , to a required output y . Here, the generator and discriminator are conditioned on any information related to the target samples, which is provided as an additional input layer in both the models. Hence, the generator is guided to produce fake samples with a specific condition rather than a generic sample from an unknown noise distribution.

The conventional CGAN was modified to suit the problem at hand. The generator synthesizes features from the fundus image, whereas the discriminator tries to distinguish between these generated features and the features obtained in the earlier stage from OCT volumes. The generator is conditioned on the input fundus image with

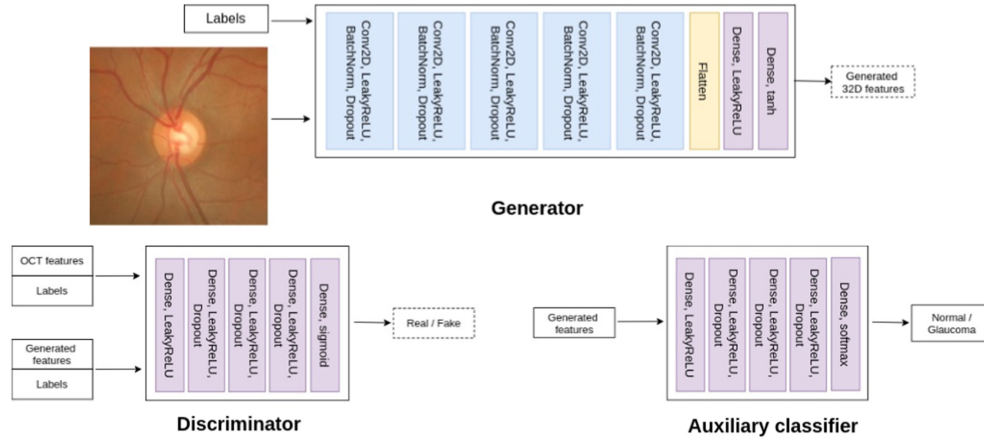


Fig. 2. Network architecture of Generator, Discriminator, and Auxiliary Classifier of the Mapping network.

the corresponding labels to produce the required feature vector. For the discriminator, feature vectors and their corresponding labels are presented as inputs. The goal is to learn a generator that maps the fundus to the OCT-feature space. An auxiliary classifier network having the same architecture as the discriminator is included, which acts as a regularizer. It takes the generated features as input and outputs the probability of the class labels. This additional classifier network is incorporated to ensure that the generated features are class specific.

The proposed network is shown in Figure 1(B) and detailed views of the network architectures are shown in Figure 2. At the end of this stage, we have a mapping from the fundus to the OCT-feature space.

Generator (G) Architecture: The generator (G) takes as input a fundus image of size 304×304 and its corresponding label and outputs a feature vector. The base architecture (B) consists of four Conv-BatchNorm-LeakyRelu-Dropout blocks and MaxPool layer after the second and fourth blocks. After flattening the output of the second MaxPool layer, a Dense layer is employed which is connected to a final Dense layer with *tanh* activation function that outputs the required feature vector. The network is regularized with a dropout of 0.4 on all the Conv and Dense layers. An Adam optimizer is used for the generator. After the training, the output feature vector obtained from G for each of the fundus inputs are used as soft labels, which are used during the training of the classification network in the next stage.

Discriminator (D) and Auxiliary classifier (C): A network consisting of four Dense layers with LeakyRelu activation functions and a dropout of 0.4 for the last two layers is employed. This base architecture is shared between the discriminator (D) and the auxiliary classifier (C) to make the training process computationally efficient. The inputs to D are the feature vectors and their corresponding labels, whereas C takes the inputs as only the feature vectors. On top of the shared network, D contains a final Dense layer with a *sigmoid* activation to determine whether the input is real or synthesized with binary cross entropy as loss function. In the case of C, two dense layers are used, where the last dense layer uses *softmax* activation to output the label by minimizing the cross entropy loss. An Adam optimizer is used for both the networks. C acts like a regularizer retaining the class information of the generated features.

2.4 Proposed Model for Classification

The overview of the model proposed for classification is illustrated in Figure 3. Since Feature fusion is the basis of the final classification result in this model, we refer to it as the **Fefu** model. A detailed view of the Fefu model is shown in Figure 4. It is a combination of three sub-networks: (i) F_{f-o} , which learns the fundus-to-OCT mapped features; (ii) F_f , which learns features specific to fundus images; and (iii) a **Fully Connected Network**

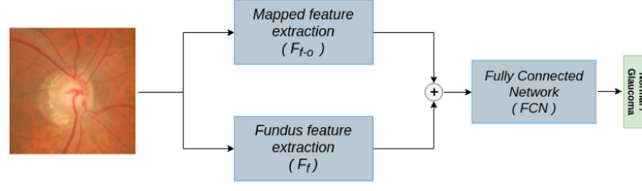


Fig. 3. Overview of the proposed classification network. F_{f-o} module is a network generating fundus-to-OCT mapped features. F_f module is a network generating fundus-specific features.

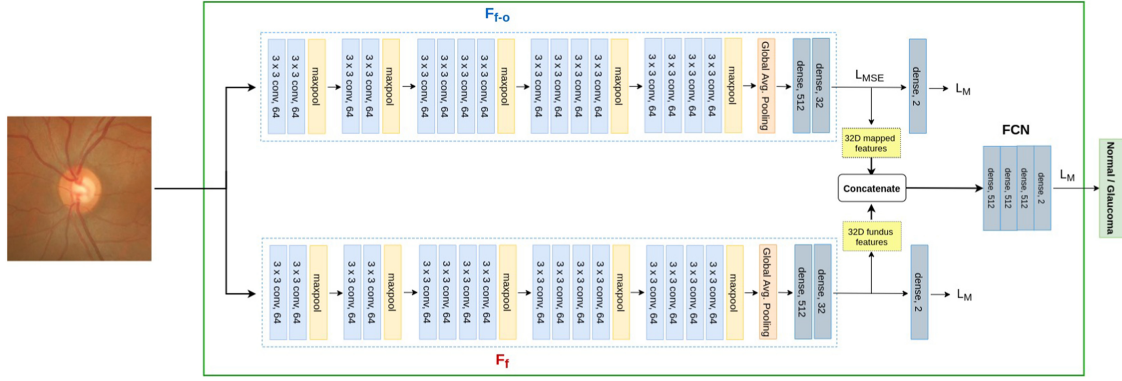


Fig. 4. Architecture of the Fefu model. L_{MSE} denotes the mean squared error loss and L_M denotes the margin loss.

(FCN), which combines the features from F_{f-o} and F_f to derive the output label. A fine-tuned VGG-19 network is employed for both F_{f-o} and F_f . The last dense layers in the original VGG-19 are replaced with **Global Average Pooling (GAP)** and three dense layers, where *softmax* activation is used in the last dense layer, as shown in Figure 4). Both F_{f-o} and F_f networks have an identical architecture to maintain symmetry and also for making the training process simpler. The feature vectors obtained from both the networks are merged and fed to the FCN, which consists of a set of four dense layers with *ReLU* activation. A probability distribution of the class label is derived by the FCN.

2.4.1 Description of F_{f-o} . The soft labels generated for fundus images (in the training set) from the previous stage are used to train this network. The main aim of this network is to generate features from the OCT-feature space along with retaining the class information from the fundus input. For this purpose, two loss functions are employed: (i) mean squared error (L_{MSE}), which minimizes the error between the generated feature vector and the soft label; and (ii) margin loss (L_M), as defined in Equation (3), which minimizes the error between the predicted and true labels. This is included to ensure the classification is robust.

2.4.2 Description of F_f . This network generates features specific to fundus images. A margin loss (L_M), as defined in Equation (3) is employed to minimize the error between the predicted and true labels.

Both F_{f-o} and F_f networks are initialized with the ImageNet weights. For the final prediction at the output of FCN, margin loss is used. All four loss functions labeled in Figure 4 are optimized in each epoch. In this way, the model incorporates both fundus and OCT image representations that are obtained from fundus images alone to improve the classification ability.

Table 1. Details of the Fundus Image Datasets

Dataset	Resolution	View	Total	Normal	Glaucoma
Private-1	$1,996 \times 1,494$	OD	1,490	706	784
Drishti-GS1	$2,896 \times 1,944$	OD	101	31	70
RIM-ONE v3	$1,072 \times 1,424$	OD	111	78	33
REFUGE	$2,124 \times 2,056$	Macula	400	360	40
sjchoi86-HRF	$2,592 \times 1,728$	Macula	401	300	101
Private-2	$4,288 \times 2,848, 2,196 \times 1,958$	OD	383	214	169

3 IMPLEMENTATION

3.1 Datasets

3.1.1 Oct. Two OCT datasets with volumes centered around OD were privately sourced. The first set of 128 volumes (95 healthy and 33 glaucomatous) of size $640 \times 304 \times 304$, were acquired with an Optovue scanner (spectral domain RTVue-XR 100 OCT). A second set of 125 volumes (69 healthy and 56 glaucomatous) of size $512 \times 512 \times 193$ were acquired with a Spectralis OCT system (Heidelberg Engineering, Germany). The two OCT datasets were combined to create a set of 253 volumes (with 164 healthy and 89 glaucomatous cases) and split randomly into three sets for training, validation, and testing. The sets had roughly a 2:1 ratio of healthy and glaucoma samples. These datasets do not contain any other retinal diseases.

3.1.2 Fundus. A total of 2,168 images were obtained by combining one private and three publicly available datasets, namely, Drishti-GS1 [30], RIM-ONE v3 [31], and REFUGE. The private dataset consisted of 1,490 *OD-centric* images along with eye/image level diagnosis. Among the public datasets, Drishti-GS1 and RIM-ONE v3 have 101 and 111 *OD-centric* images, respectively, whereas REFUGE has 400 *macula-centric* images. The REFUGE dataset was released as part of a challenge [32]. The distribution of normal and glaucomatous cases in these datasets is given in Table 1. It can be noted from this table that the image sources, acquisition protocol (view, resolution) vary widely. Hence, these were combined to create a training and a testing dataset. The contribution of the four datasets to the training and testing sets is shown in Table 2. It can be observed from Table 1 that a major proportion of the images are OD-centric as it captures the structures (OD and OC) and the peripapillary regions without any distortion.

Evaluation of the network on unseen data was carried out on two more datasets (last two rows of Table 1): A public dataset, namely, sjchoi86-HRF, was obtained from [33] which consists of *macula-centric* images and a locally sourced (Private-2) dataset consisting of *OD-centric* images acquired with two different fundus cameras. In order to handle the wide variations in image resolution, all the images were preprocessed to a common format using the same process as described in 2.2.

The combined dataset consisting of 2,168 images was randomly split into training and testing sets in our experiments. Following the conventional method, diverse datasets were combined to ensure the proposed model is unbiased. The distribution of train, validation, and test sets for the capsule networks, CGAN, and Fefu is provided in Table 3.

3.2 Experimental Setting

3.2.1 OCT Capsule Network. The 3D capsule network architecture consists of 243,736 network parameters. The number of iterations for routing is a hyperparameter for the Capsule network and was chosen to be three since larger values might lead to overfitting. Fivefold cross-validation was done for 50 epochs using an Adam optimizer with a learning rate of 0.0001 which decays at the rate of 0.9 for faster training. The batch size was set to 3 because of memory constraints. The class imbalance in the dataset was handled by embedding class weights in the loss function.

Table 2. Decomposition of the Combined Set Used for the Fefu Model's Training/Testing in Terms of the Individual Datasets

	Normal (Train/Test)	Glaucoma (Train/Test)
Private-1	430/299	715/112
Drishti-GS1	20/11	40/30
RIM-ONE v3	50/28	20/13
REFUGE	300/60	25/15
Total	800/398	800/170

Table 3. Distribution of Datasets for Training/Testing/Validation of the Different Modules

	Train (N/G)	Test (N/G)	Validation (N/G)
Capsule network (OCT)	165 (106/59)	38 (26/12)	50 (32/18)
CGAN (Fundus)	1,278 (605/673)	212 (101/111)	-
Fefu (Fundus)	1,440 (720/720)	568 (398/170)	160(80/80)

N and G are the number of normal and glaucomatous samples.

Table 4. Performance of the Capsule Network (OCT)

	BAcc.	Sens.	Spec.	AUC
OCT-Capsule network	0.92	0.89	0.96	0.97

3.2.2 Fundus Networks. The mapping architecture (CGAN) consists of 1,604,639 parameters and was trained for 700 epochs using an Adam optimizer with a batch size of 100, learning rate of 0.001, and decaying at the rate of 0.8. F_f and F_{f-o} have 20,040,866 parameters each and the FCN network consists of 559,874 parameters. The train set was passed through the CGAN to obtain their respective soft-labels (feature vector) for the F_{f-o} module of the Fefu model. The Fefu model was trained for 100 epochs using a SGD optimizer with a learning rate of 0.001, momentum of 0.9, decay rate of 1e-8, and a batch size of 32 with fivefold cross-validation. Data augmentation is done by random flips and random rotations by 0/90/180/270 degrees. Quantitative evaluation for all the stages was done using the following metrics: Balanced Accuracy (BAcc.), Sensitivity (Sens.), Specificity (Spec.), and AUC.

$$Sens. = \frac{TP}{TP + FN}, \quad Spec. = \frac{TN}{TN + FP}, \quad BAcc. = \frac{Sens. + Spec.}{2}, \quad (5)$$

where TP, TN, FP, and FN denote the number of true positives, true negatives, false positives, and false negatives, respectively. Along with these, ROC plots were also derived. The optimum threshold was calculated by using the Youden index [34], where the sum of the Specificity-1 and Sensitivity is maximum. All frameworks were implemented on Keras with Tensorflow backend on a NVIDIA GTX 1080 GPU, with 11 GB of GPU RAM.

4 EXPERIMENTS AND RESULTS

The 3D Capsule network, which extracts features from OCT volumes, was tested on a set of 50 volumes. The performance of the network is reported in Table 4. A high value for AUC (0.97) indicates that the Capsule network has learned highly discriminative features from the 3D OCT volumes. The capsule features of the correctly predicted volumes (233 = 153 normal + 80 glaucomatous) are passed on to the mapping network (the block labeled as **B** in Figure 1).

The correlation between the OCT-features and mapped features was also assessed with a set of 112 paired OCT, and fundus images were used to derive the OCT-features (from the capsule network) and the mapped fundus

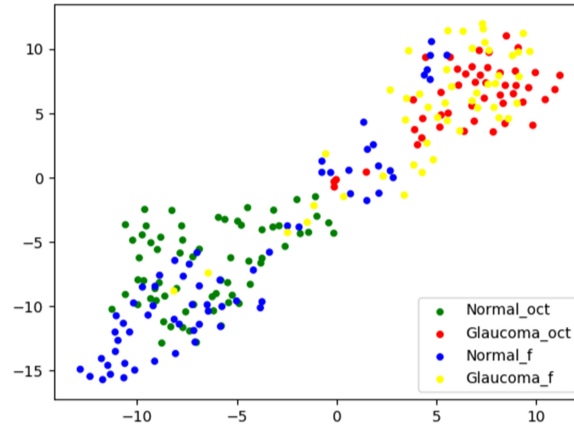


Fig. 5. t-SNE visualization of features. The red and green represent the OCT-features. The blue and yellow represent features obtained from the mapped network (F_{f-o}).

Table 5. Performance of the Proposed Fefu Model on the Combined Test Set

	BAcc.	Sens.	Spec.	AUC
Only F_f	0.8252	0.8573	0.7930	0.8882
Only F_{f-o}	0.8661	0.8809	0.8514	0.9240
Fefu model (Proposed)	0.8909	0.9044	0.8773	0.9429

features ($f - o$). To visually assess the correlation, t-SNE was used to convert the feature vectors to 2D, which are plotted in Figure 5. It can be observed that the healthy and glaucoma classes can be differentiated reasonably well in the mapped feature space.

The proposed system was assessed extensively, which is described next.

4.1 Performance on the Combined Dataset

A fivefold cross-validation was performed to assess the Fefu model in Figure 3 using the combined test set consisting of 568 images (328 normal and 170 glaucomatous). The average value for each of the metrics is reported in Table 5 and the corresponding ROC curves are shown in Figure 6. The individual contribution of the two networks F_{f-o} and F_f was also assessed to gauge the level of contributions of the Fundus (f) only and Fundus to OCT ($f - o$) mapped features. The two networks were trained for 50 epochs individually on the combined training set for this study. In both these cases, fivefold cross-validation was performed. The tabulated results indicate the improvement in AUC (with f as baseline) is 4% with mapped features ($f - o$) and 6% with combined features. Hence, it can be concluded that mapped features are superior to unmapped features for classification of an image as being normal or glaucomatous, and combining features is beneficial.

4.2 Performance on Individual Datasets

The results reported in Table 6 are based on a detailed analysis of the performance of the Fefu model and its modules on individual datasets that constituted the combined test dataset. The Sens. and Spec. figures are reported (as before) at the optimum threshold. This is to assess the effect of trained networks on individual datasets. Overall, it can be noted that the Fefu model has consistently high performance ($0.85 \leq BAcc. \leq 0.96$; $0.91 \leq AUC \leq 0.99$) on all datasets, which is much higher than that of F_f ($0.76 \leq BAcc. \leq 0.85$; $0.82 \leq AUC \leq 0.90$). It is also worth

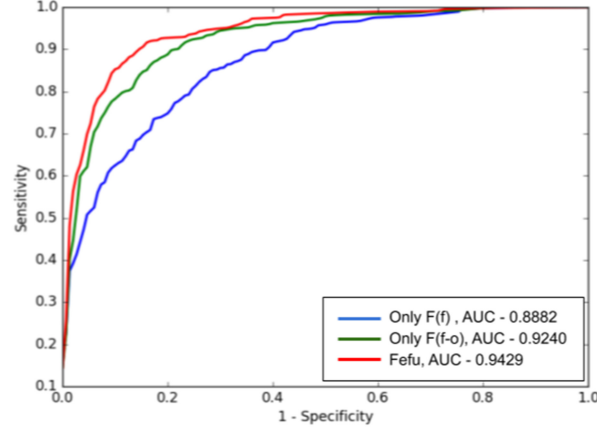


Fig. 6. ROC curves along with AUC comparing the performance of different networks on the combined test set corresponding to Table 5.

Table 6. Performance of the Proposed Fefu Model on the Individual Test Sets

	Private-1 (N = 299, G = 112)				Drishti-GS1 (N = 11, G = 30)			
	B Acc.	Sens.	Spec.	AUC	B Acc.	Sens.	Spec.	AUC
Only F_f	0.7578	0.7649	0.7507	0.8204	0.8258	0.8333	0.8182	0.9030
Only F_{f-o}	0.8021	0.8175	0.7867	0.8633	0.8997	0.9267	0.8698	0.9473
Fefu model	0.8511	0.8860	0.8162	0.9067	0.9167	0.8333	1.0000	0.9667
	RIM-ONE v3 (N = 28, G = 13)				REFUGE (N = 60, G = 15)			
	B Acc.	Sens.	Spec.	AUC	B Acc.	Sens.	Spec.	AUC
Only F_f	0.8489	0.7692	0.9286	0.9011	0.8333	0.8000	0.8667	0.8878
Only F_{f-o}	0.9466	0.9077	0.9857	0.9664	0.9250	0.8800	0.9700	0.9346
Fefu model	0.9643	1.0000	0.9286	0.9945	0.9333	0.8667	1.0000	0.9100

N and G denote the number of normal and glaucomatous samples, respectively.

noting that there is a 5–7% improvement in the performance of F_{f-o} relative to F_f for all the datasets, implying a superiority of the Fundus-OCT feature space over the fundus-feature space for glaucoma assessment.

4.3 Comparison with State-of-the-Art Method

The proposed system was cross validated and compared with DE-Net [15], a state-of-the-art method. DE-Net derives the classification decision by combining the predictions of four independent neural network streams using ResNet-50. The pre-trained weights for the DE-Net available online [35] were used in this experiment.

We begin with cross-validation results of the DE-Net. The $B\text{Acc}/AUC$ was found to be 0.67/0.72 for Private-1, 0.83/0.85 for Drishti-GS1, 0.72/0.78 for RIM-ONE v3, and 0.76/0.83 for REFUGE datasets. These cannot be directly compared with the figures in Table 6 as the Fefu model was trained on a combined set generated by splitting these datasets. Nevertheless, the figures exhibit a downward trend. For a fairer comparison, two independent (i.e., neither were part of the combined set used for training the proposed networks) datasets were chosen for cross-validation: a public (sjchoi86-HRF) dataset and a locally sourced one named as Private-2. The obtained results presented in Table 7 show that the proposed model outperforms the DE-Net on both the unseen datasets. Specifically, in terms of AUC , the difference is 18.6% on the Private-2. Relative to this, the proposed model's performance on sjchoi86-HRF is lower, perhaps due to the fact that nearly 80% of the training images for our

Table 7. Cross-Validation Performance Comparison of Different Networks on the Individual Test Sets

	Private-2 (N = 214, G = 169)				sjchoi86-HRF (N = 300, G = 101)			
	B Acc.	Sens.	Spec.	AUC	B Acc.	Sens.	Spec.	AUC
Only F_f	0.7675	0.7219	0.8131	0.8356	0.6840	0.6700	0.6980	0.7290
Only F_{f-o}	0.8323	0.8047	0.8598	0.9048	0.7358	0.7400	0.7315	0.7841
Fefu model	0.8707	0.8817	0.8598	0.9162	0.7558	0.7700	0.7416	0.8089
DE-Net (2018) [15]	0.7182	0.6607	0.7757	0.7724	0.6997	0.7327	0.6667	0.7490
MNet (2018) [13]	0.3655	0.3495	0.3846	-	0.6992	0.7018	0.6915	-

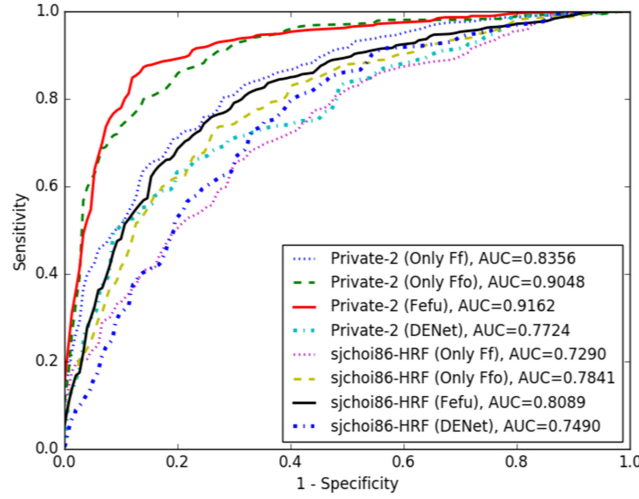


Fig. 7. ROC curves for cross-validation performance of different networks, corresponding to the first two rows of Table 7.

model were OD-centric, whereas the sjchoi86-HRF has only macula-centric images. However, our model's AUC is still 8% higher than that of DE-Net. It is also noteworthy that the AUC values obtained for F_{f-o} are 17.1% and 4.6% higher than DENet on the Private-2 and sjchoi86-HRF. This underscores the superiority of the OCT-Fundus feature space over feature space obtained using only fundus images.

Our model was also compared against CDR-based classification as it is popular for glaucoma. The pre-trained MNet [13] model (available online) was used to derive the CDR. Next, the **International Classification of Diseases (ICD-9)** rule on CDR was used to classify the images into two classes: normal ($0 \leq CDR < 0.5$) and glaucomatous ($0.5 \leq CDR \leq 1$). The results reported in Table 7 and the corresponding ROC curves in Figure 7 indicate our model to be superior.

4.4 Visualization

Class Activation Mapping (CAM) is a visualization technique that allows us to understand what networks look at when they classify an input image, thus making it possible to visually understand why the network chooses one class over another. More recently, improving upon CAM, **Gradient-weighted Class Activation Mapping (Grad-CAM)** [36] is introduced. In particular, Grad-CAM uses class-specific gradient information flowing into the final convolutional layer of the network to produce a coarse localization map of the important regions in a given input image. Grad-CAMs were computed for F_f and F_{f-o} networks, resized and overlayed on the input fundus, to identify the regions that the networks sees to be important for the classification decision. Figures 8–11 depict the Grad-CAMs for few glaucoma samples from Private-2 and sjchoi86-HRF sets. As

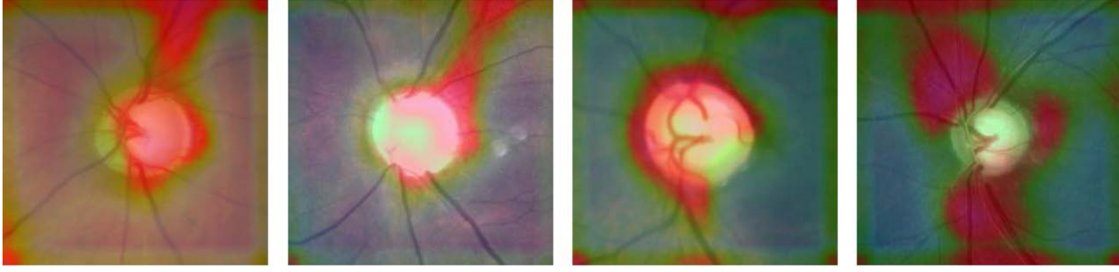


Fig. 8. Grad-CAM visualization of glaucoma samples from the Private-2 set that are predicted by F_f .

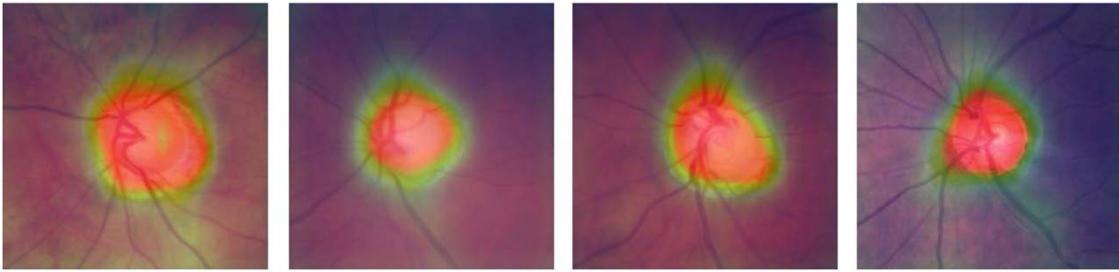


Fig. 9. Grad-CAM visualization of glaucoma samples from the Private-2 set that are predicted by F_{f-o} .

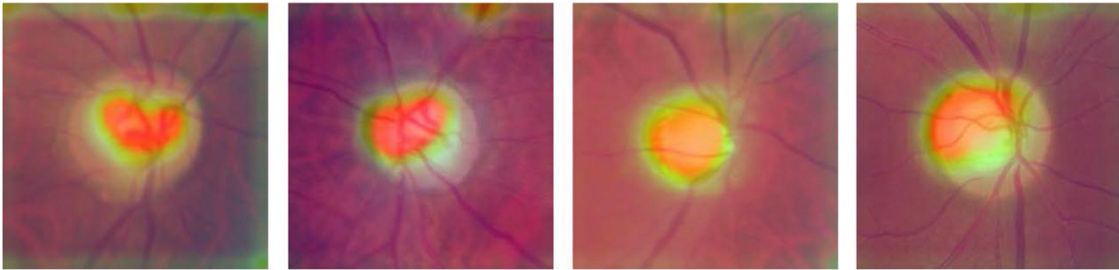


Fig. 10. Grad-CAM visualization of glaucoma samples from the sjchoi86-HRF set that are predicted by F_f .

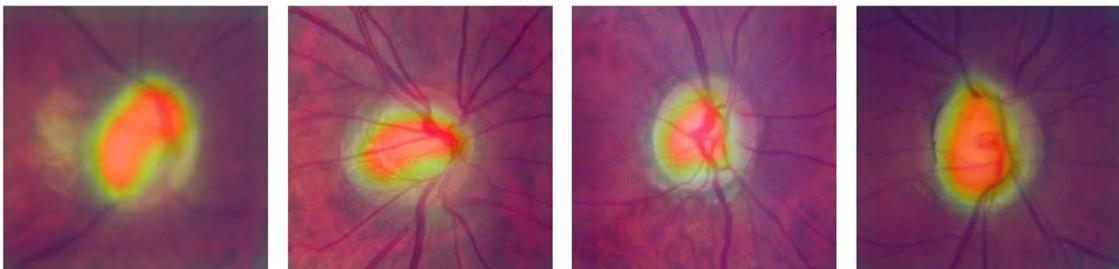


Fig. 11. Grad-CAM visualization of glaucoma samples from the sjchoi86-HRF set that are predicted by F_{f-o} .

illustrated in the localization maps of the glaucoma samples, it can be observed from Figures 8 and 10 that F_f seems to identify the cup and superior and inferior RNFL regions, whereas in the case of F_{f-o} it is the disc region (Figures 9 and 11). These regions correspond well to the common biomarkers for glaucoma diagnosis.

5 CONCLUSION

We presented a solution for glaucoma assessment which utilizes unpaired data for mapping fundus to OCT feature space, hence bridging the gap between the two modalities. Thus, in our method, the OCT and fundus need not be paired, which is one of the key advantages, as sourcing paired data is difficult and expensive. It is noteworthy from the results that the fundus-OCT feature space has more discriminative ability compared to the fundus-feature space, which was further boosted on the combination of both. Another important point to note is that a strong and consistent (across datasets) performance can be obtained with a lighter network with fewer parameters by using such mapping approach and that doesn't rely on segmentation. This is in contrast to existing methods (that rely only on fundus images) which employ multiple networks and a large number of parameters. Smaller networks are attractive in screening scenarios as the online computing can then be done on the imaging device (such as a smartphone-based fundus imaging system) itself. The proposed idea was implemented largely with conventional networks (capsule, CGAN, VGG, etc.) without any need for segmentation of structures. Hence, in the future, other custom designs for the networks can be explored to implement the proposed ideas as well. Further, variations of the proposed model can be investigated with additional inputs like patient's meta information after getting access to such data in the future.

ACKNOWLEDGMENTS

We would like to thank the following for providing access to their study data: Dr. Krishnadas, Aravind Eye Hospital, India; Dr. T. Mansoori, Anand Eye Hospital, India; and Dr. K. A. Vermeer, Dr. H. G. Lemij, and J. E. A. Majoor from the Rotterdam Eye Hospital, The Netherlands.

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Received May 2020; revised February 2021; accepted June 2021